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4. Covariance and Correlation Matrix

- *Variance of the data X can be defined as follows : (As you see, in case of single column data, Variance is same as covariance between the same data).*
- *Now let's assume that we have two set of single column data as shown below.*

Deviation from the Mean of X

Mean of X

$$Var(X) = \frac{\sum (x_i - \bar{X})(x_i - \bar{X})}{N}$$

$$= \frac{\sum (x_i - \bar{X})^2}{N}$$

$$= \text{Cov}(X, X)$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix}$$

Deviation from the Mean of X

Mean of X

$$Var(X) = \frac{\sum (x_i - \bar{X})^2}{N}$$

Deviation from the Mean of Y

Mean of Y

$$Var(Y) = \frac{\sum (y_i - \bar{Y})^2}{N}$$

4. Covariance and Correlation Matrix

- *The covariance between these two data set (two single column data) is defined as follows :*

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix}$$

Deviation from the Mean of X — Deviation from the Mean of Y

$$\mathrm{Cov}(X,Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{N}$$

- *If you combine the two sets of data into a matrix form as shown below*

$$A = \begin{matrix} X & Y \\ \downarrow & \downarrow \end{matrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_{n-1} & y_{n-1} \\ x_n & y_n \end{bmatrix}$$

$$\mathrm{Cov}(A) = \begin{bmatrix} \frac{\sum (x_i - \bar{X})(x_i - \bar{X})}{N} & \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{N} \\ \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{N} & \frac{\sum (y_i - \bar{Y})(y_i - \bar{Y})}{N} \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{Cov}(X,X) & \mathrm{Cov}(Y,X) \\ \mathrm{Cov}(X,Y) & \mathrm{Cov}(Y,Y) \end{bmatrix}$$

4. Covariance and Correlation Matrix

- *Properties of Covariance Matrix*
  - *Covariance matrix can represents the variance and linear correlation in multivariate/multidimensional data*
  - *Covariance matrix gives you meaningful result only when the data sets have linear correlation.*
  - *Covariance matrix is always a Square Matrix*
  - *If Idata is n rows and m columns (n x m Matrix), the dimension of Covariance matrix is m x m Square matrix*
- *Covariation vs. Correlation*

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$$

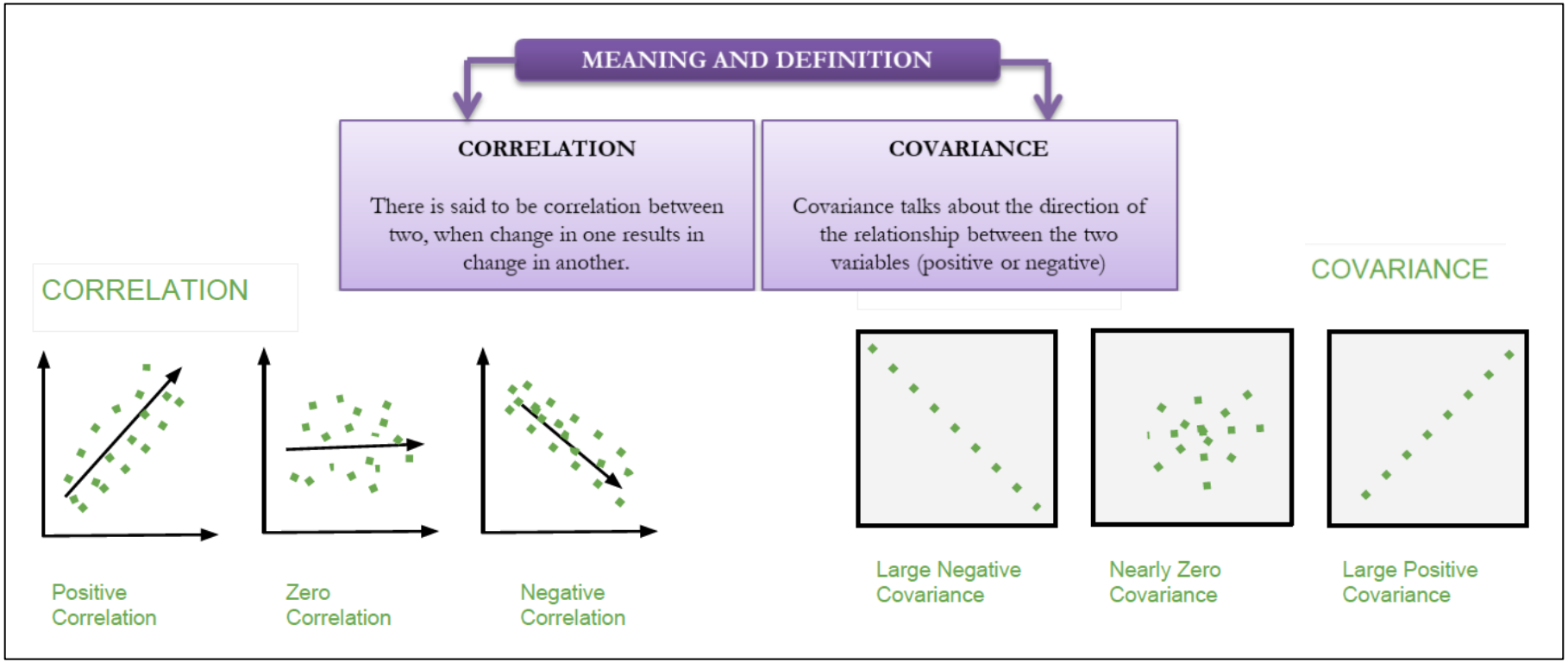
Covarianced normalized by Standard Deviation

Correlation between X and Y

Standard deviation of X

Standard deviation of Y

4. Covariance and Correlation Matrix

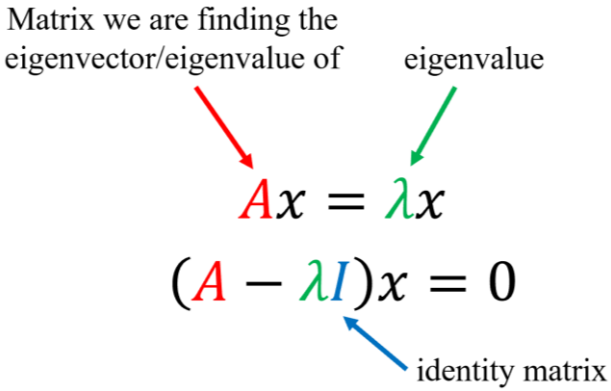


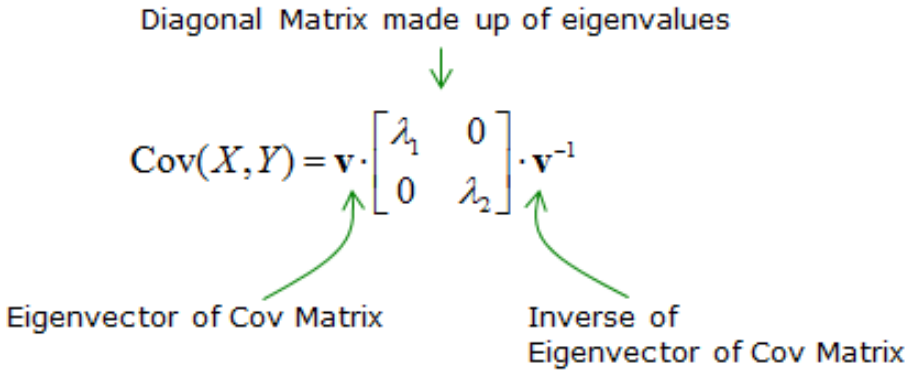
4. Covariance and Correlation Matrix

CORRELATION	COVARIANCE
Measures the strength of the variables under comparison	Measures the extent of change in one with regards to change in another.
Correlation is a scaled down version of covariance.	Covariance is considered as a part of correlation.
Value here lies between -1 and +1.	Value here lies between -infinity to +infinity
Correlation is a unit-free measure	Covariance value is the product of the units of the variables.
There would be no change in correlation due to scale.	Any change in scale affects covariance.

5. Eigen Values and Eigen Vectors

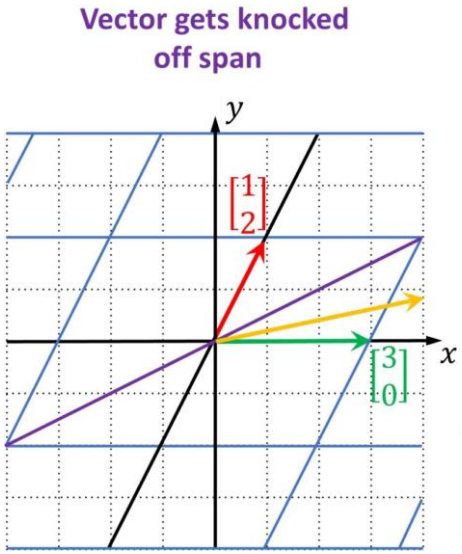
- *In linear algebra, an eigenvector or characteristic vector of a linear transformation is a nonzero vector that changes at most by a scalar factor when that linear transformation is applied to it.*
- *The corresponding eigenvalue is the factor by which the eigenvector is scaled.*





5. Eigen Values and Eigen Vectors

Such transformation for those λ will squish the plane into a single line

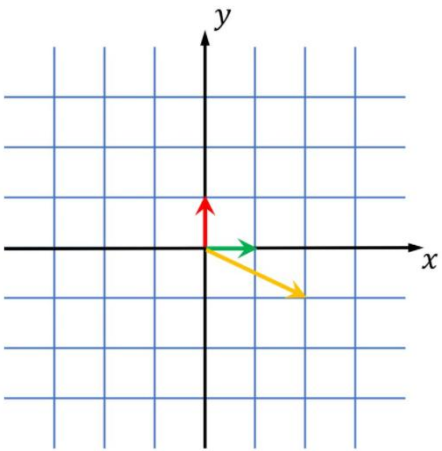


$$A\vec{v} = \lambda\vec{v}$$
$$A\vec{v} - \lambda I\vec{v} = 0$$
$$(A - \lambda I)\vec{v} = 0$$
$$\det(A - \lambda I) = 0$$

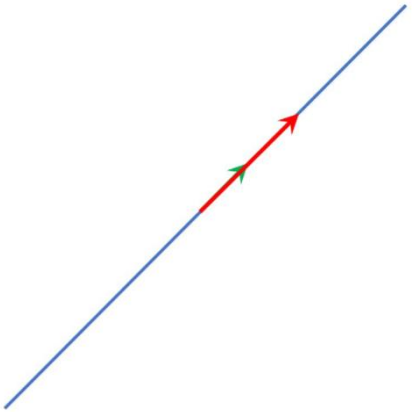
$$(A - \lambda I)\vec{v} = \vec{0}$$

$$\begin{bmatrix} 3-\lambda & 1 & 4 \\ 1 & 5-\lambda & 9 \\ 2 & 6 & 5-\lambda \end{bmatrix} = 0$$

$(A - \lambda I)\vec{v} = \vec{0}$



$\det(A - \lambda I)\vec{v} = 0$



5. Eigen Values and Eigen Vectors

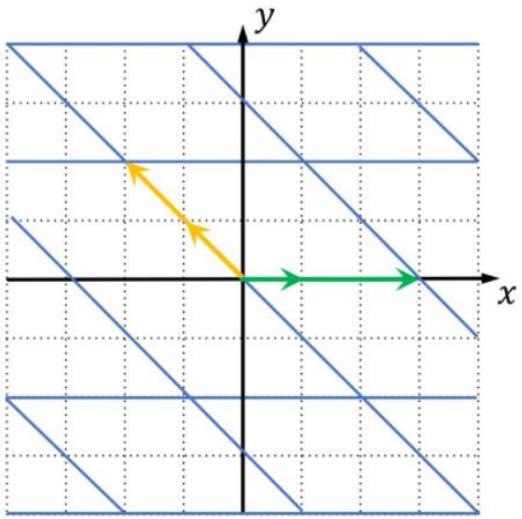
$$\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}^{-1}\begin{bmatrix}3 & 1\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}$$

$$\det\left(\begin{bmatrix}3-\lambda & 1\\ 0 & 2-\lambda\end{bmatrix}\right)=(3-\lambda)(2-\lambda)$$

Quadratic polynomial in λ

$$\det\left(\begin{bmatrix}3-\lambda & 1\\ 0 & 2-\lambda\end{bmatrix}\right)=(3-\lambda)(2-\lambda)$$

$$\lambda=2 \quad \text{or} \quad \lambda=3$$

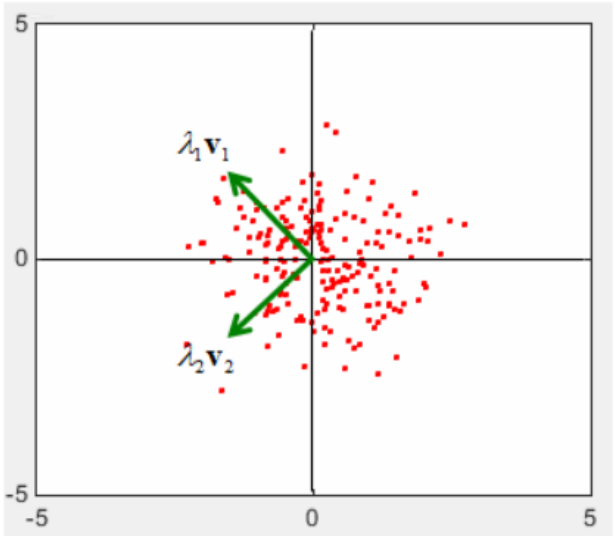


We know that everything along these vectors will be just multiplied with eigenvalues associated with these eigenvectors.
So, along these directions the changes will be just multiplied and they will not change the direction.

$$\begin{bmatrix}3 & 1\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}$$

5. Eigen Values and Eigen Vectors

Example 1



$$\text{Cov}(X,Y) = \begin{bmatrix} 1.0020 & -0.0978 \\ -0.0978 & 1.0127 \end{bmatrix}$$

Eigenvector of Cov Matrix

$$\mathbf{v} = \begin{bmatrix} -0.6875 & -0.7262 \\ 0.7262 & -0.6875 \end{bmatrix}$$

$\mathbf{v}_1$ $\mathbf{v}_2$

$$\lambda_1 = 1.1053$$
$$\lambda_2 = 0.0904$$

Eigenvalues of Cov Matrix

Diagonal Matrix made up of eigenvalues

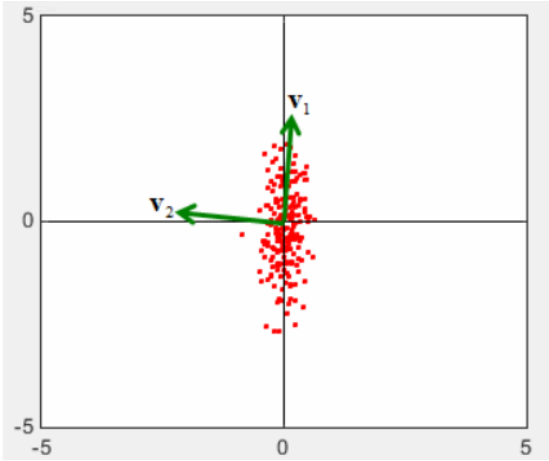
$$\text{Cov}(X,Y) = \mathbf{v} \cdot \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \cdot \mathbf{v}^{-1}$$

Eigenvector of Cov Matrix

Inverse of Eigenvector of Cov Matrix

5. Eigen Values and Eigen Vectors

Example 2



$$\text{Cov}(X,Y) = \begin{bmatrix} 0.0601 & 0.0268 \\ 0.0268 & 0.9444 \end{bmatrix}$$

Eigenvector of Cov Matrix

$\mathbf{v}_1$ $\mathbf{v}_2$

$$\mathbf{v} = \begin{bmatrix} 0.0302 & -0.9995 \\ 0.9995 & 0.0302 \end{bmatrix}$$

$\lambda_1 = 0.9452$

$\lambda_2 = 0.0593$

Eigenvalues of Cov Matrix

Diagonal Matrix made up of eigenvalues

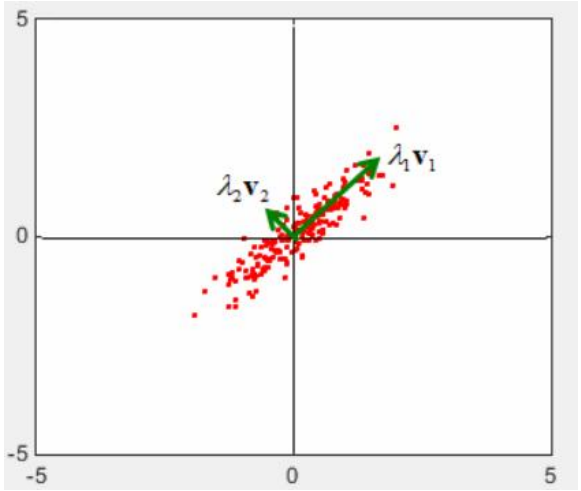
$$\text{Cov}(X,Y) = \mathbf{v} \cdot \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \cdot \mathbf{v}^{-1}$$

Eigenvector of Cov Matrix

Inverse of Eigenvector of Cov Matrix

5. Eigen Values and Eigen Vectors

Example 3



$$\text{Cov}(X,Y) = \begin{bmatrix} 0.5592 & 0.5071 \\ 0.5071 & 0.5628 \end{bmatrix}$$

Eigenvector of Cov Matrix

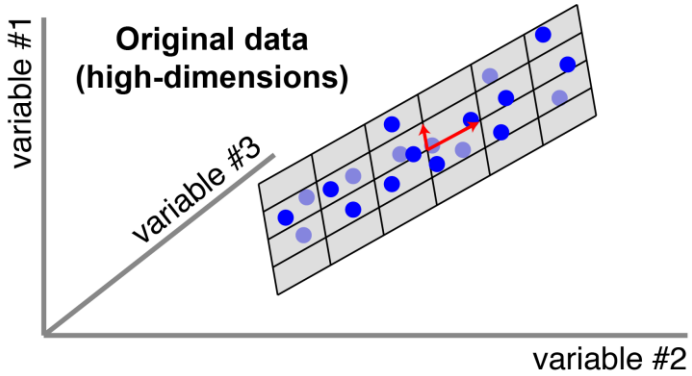
$$\mathbf{v} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \\ 0.7058 & -0.7084 \\ 0.7084 & 0.7058 \end{bmatrix}$$

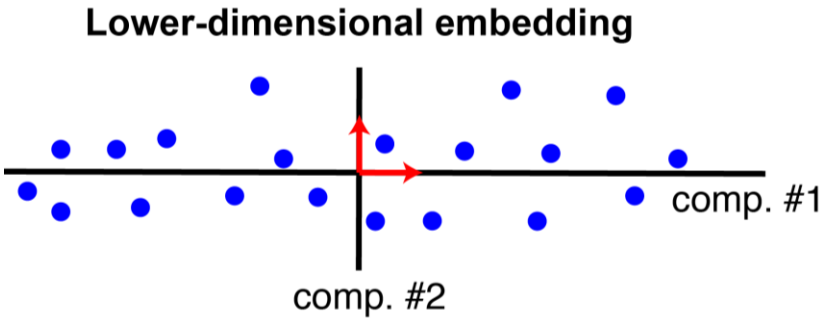
$$\lambda_1 = 1.0681$$
$$\lambda_2 = 0.0538$$

Eigenvalues of Cov Matrix

6. Principle Component Analysis

- *Principal Component Analysis, or PCA, is a dimensionality-reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set.*
- *Because smaller data sets are easier to explore and visualize and make analyzing data much easier and faster for machine learning algorithms without extraneous variables to process.*





6. Principle Component Analysis

- *STEP BY STEP EXPLANATION OF PCA*
- *STEP 1: STANDARDIZATION*
  - *The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.*
  - *Mathematically, this can be done by subtracting the mean and dividing by the standard deviation for each value of each variable.*

$$z = \frac{value - mean}{standard\ deviation}$$

- *STEP 2: COVARIANCE MATRIX COMPUTATION*
  - *The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other, or in other words, to see if there is any relationship between them*

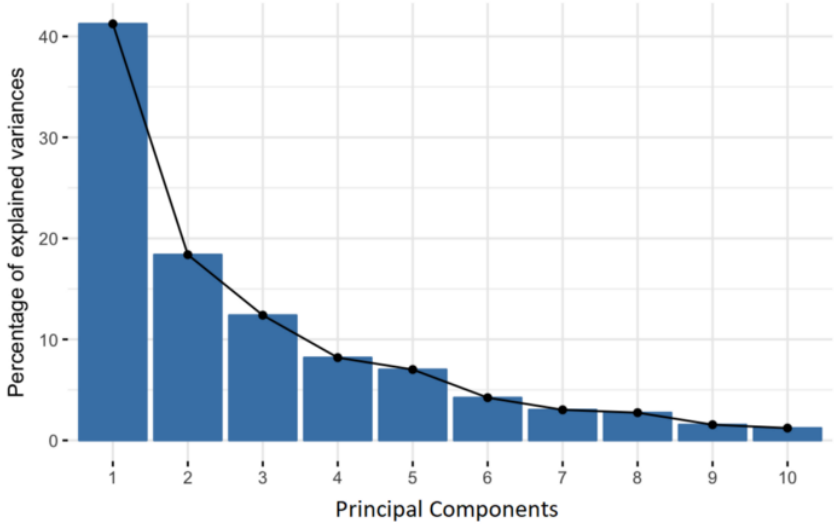
$$\begin{bmatrix} Cov(x,x) & Cov(x,y) & Cov(x,z) \\ Cov(y,x) & Cov(y,y) & Cov(y,z) \\ Cov(z,x) & Cov(z,y) & Cov(z,z) \end{bmatrix}$$

6. Principle Component Analysis

STEP BY STEP EXPLANATION OF PCA

STEP 3: COMPUTE THE EIGENVECTORS AND EIGENVALUES OF THE COVARIANCE MATRIX TO IDENTIFY THE PRINCIPAL COMPONENTS

- *Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data.*
- *Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables*
- *Most of the information within the initial variables is squeezed or compressed into the first components*



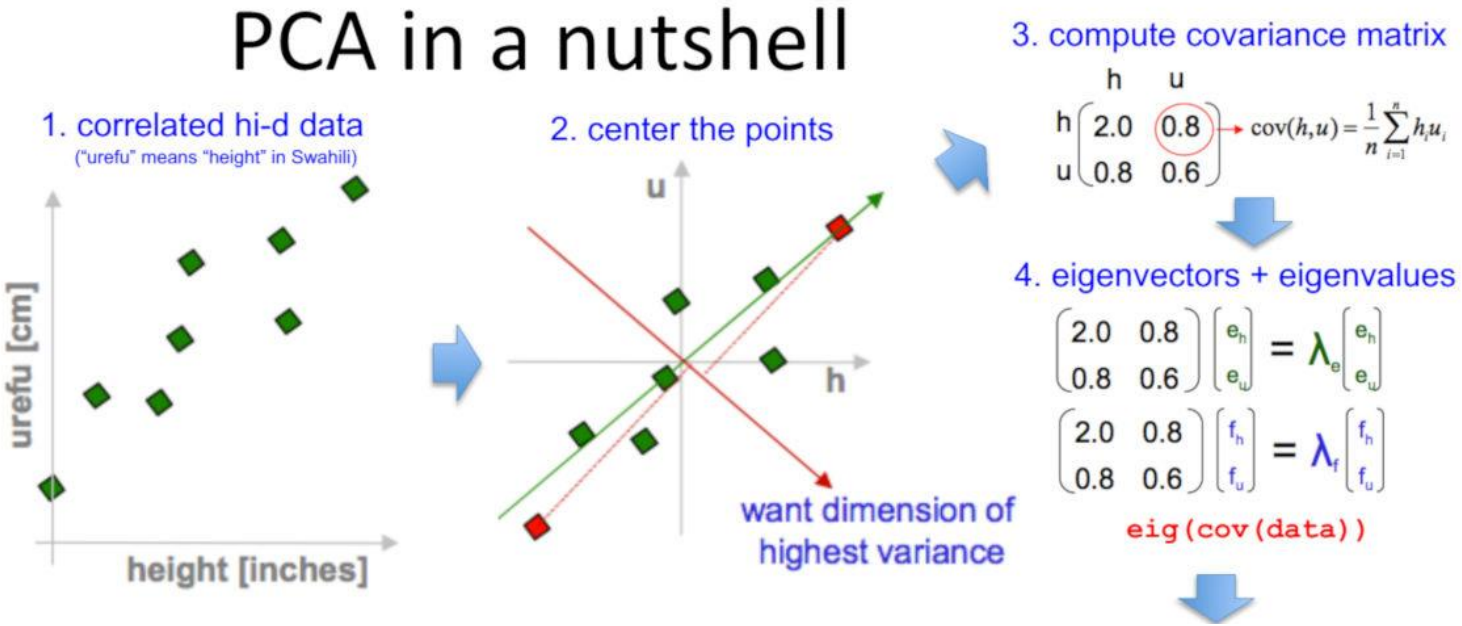
6. Principle Component Analysis

- *STEP BY STEP EXPLANATION OF PCA*
- *STEP 4: FEATURE VECTOR*
  - *Computing the eigenvectors and ordering them by their eigenvalues in descending order, allow us to find the principal components in order of significance.*
  - *In this step, what we do is, to choose whether to keep all these components or discard those of lesser significance (of low eigenvalues), and form with the remaining ones a matrix of vectors that we call Feature vector.*
- *LAST STEP: RECAST THE DATA ALONG THE PRINCIPAL COMPONENTS AXES*
  - *In this step, which is the last one, the aim is to use the feature vector formed using the eigenvectors of the covariance matrix, to reorient the data from the original axes to the ones represented by the principal components*
  - *This can be done by multiplying the transpose of the original data set by the transpose of the feature vector.*

$$FinalDataSet = FeatureVector^T * StandardizedOriginalDataSet^T$$

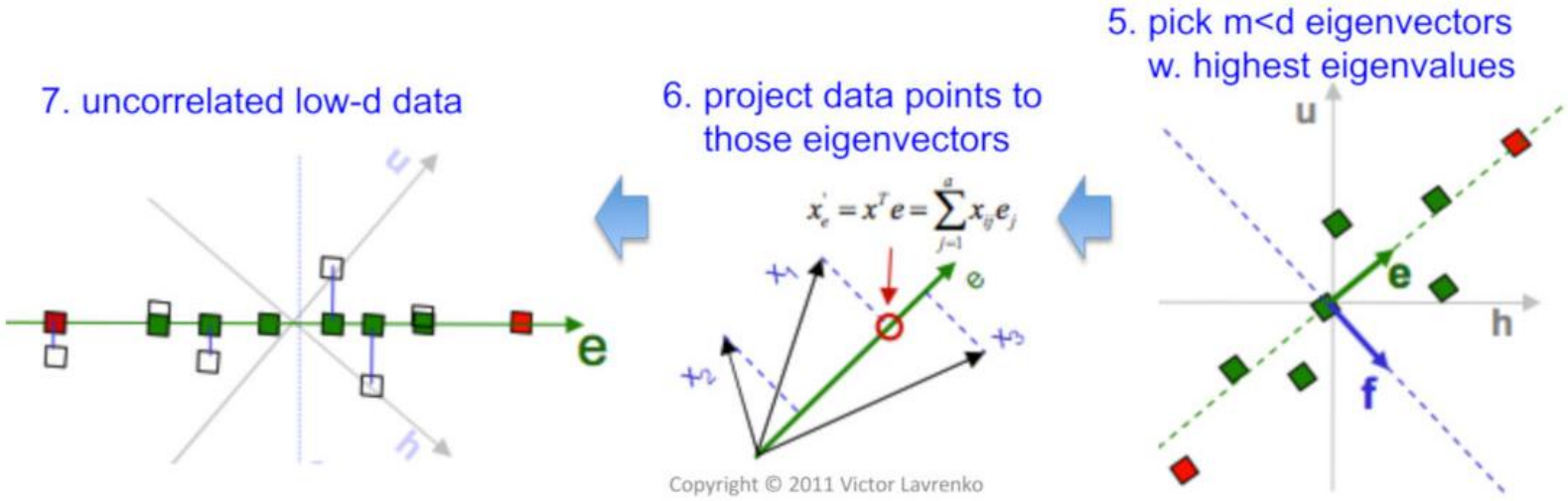
6. Principle Component Analysis

Example



6. Principle Component Analysis

Example



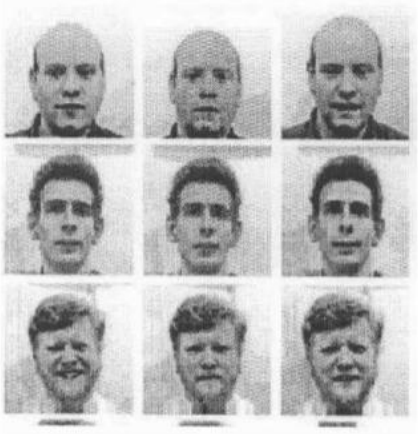
6. Principle Component Analysis

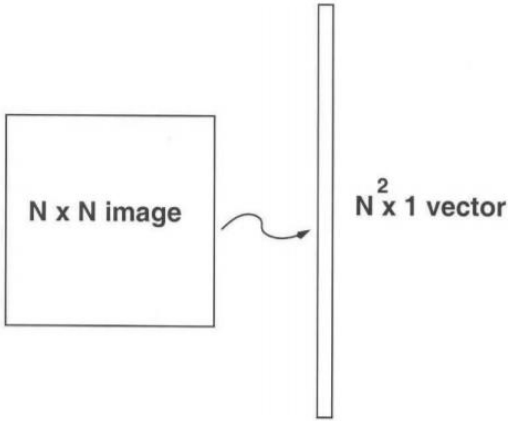
Example

Computation of the eigenfaces

Step 1: obtain face images $I_1, I_2, ..., I_M$ (training faces)

(**very important:** the face images must be *centered* and of the same *size*)





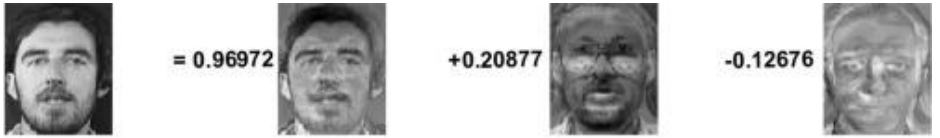
Step 2: represent every image I_i as a vector Γ_i

6. Principle Component Analysis

Example

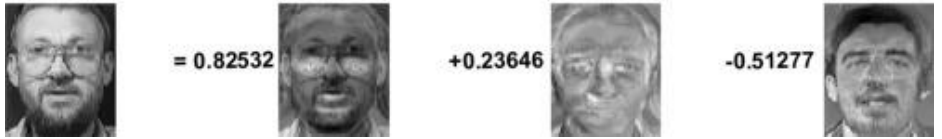
Step 3: compute the average face vector Ψ:

$$\Psi = \frac{1}{M}\sum_{i=1}^{M}\Gamma_i$$



Step 4: subtract the mean face:

$$\Phi_i = \Gamma_i - \Psi$$



Step 5: compute the covariance matrix C:

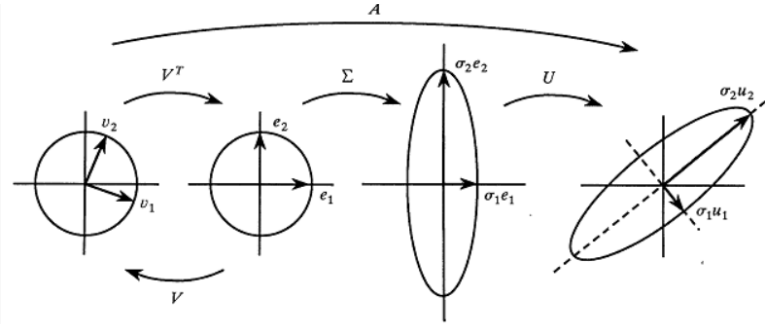
$$C = \frac{1}{M}\sum_{n=1}^{M}\Phi_n\Phi_n^T = AA^T \quad (N^2 \text{x} N^2 \text{ matrix})$$

where $A = [\Phi_1\ \Phi_2 \cdots \Phi_M]$ $\quad (N^2 \text{x} M \text{ matrix})$

7. Singular Value Decomposition

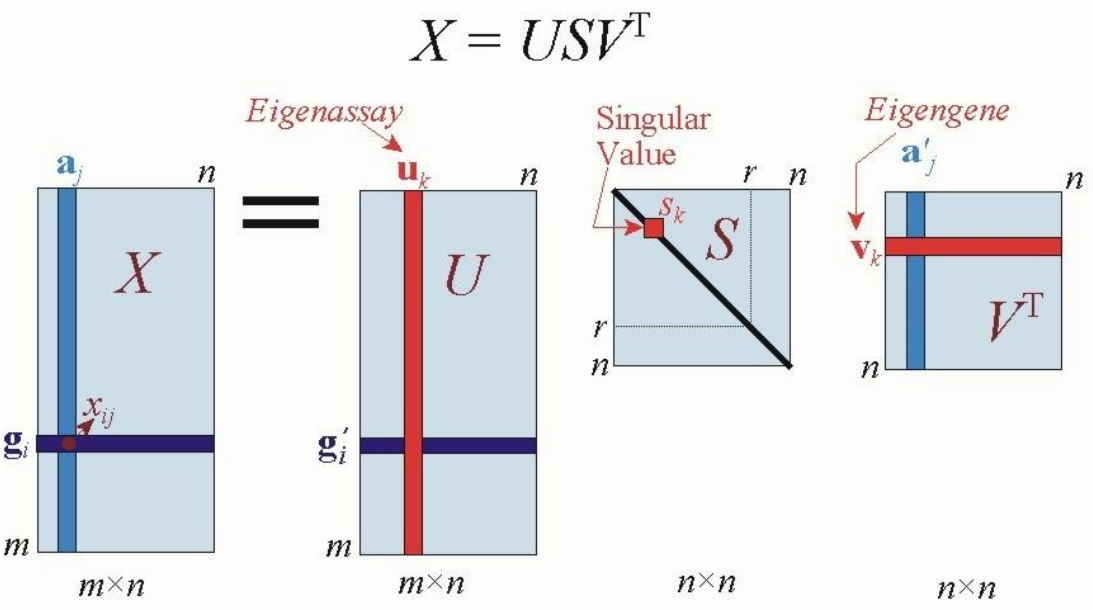
- *The SVD represents the essential geometry of a linear transformation. It tells us that every linear transformation is a composition of three fundamental actions. Reading the equation from right to left:*
  - *The **matrix V** represents a **rotation or reflection of vectors** in the p-dimensional domain.*
  - *The **matrix Σ** represents a **linear dilation or contraction** along each of the p coordinate directions. If n ≠ p, this step also canonically embeds (or projects) the p-dimensional domain into (or onto) the n-dimensional range.*
  - *The **matrix U** represents a **rotation or reflection of vectors** in the n-dimensional range.*

$$A = U \Sigma V^T$$



7. Singular Value Decomposition

Example



7. Singular Value Decomposition

Example

$$\mathbf{A} = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ a_1 & a_2 & \cdots & a_m \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$\mathbf{A_k} \in \mathbb{R}^n$



SVD: The fundamental theorem of multivariate data analysis. Because of its usefulness, the singular value decomposition is a fundamental technique for multivariate data analysis.

** Low-rank approximations*

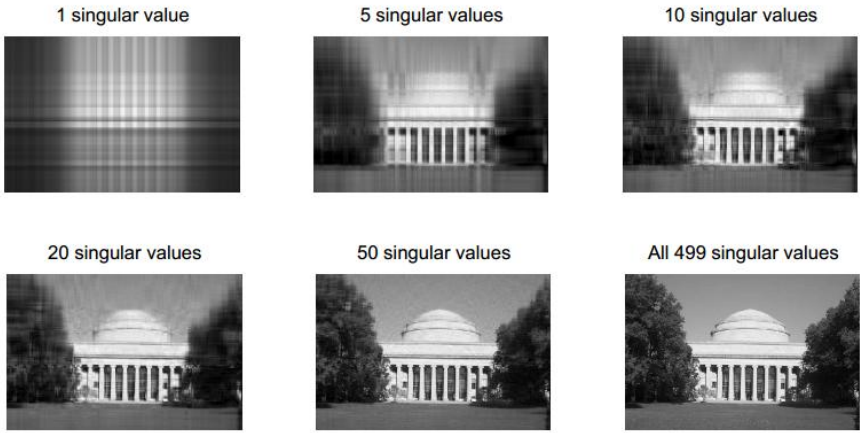
** Principal components analysis*

$$\mathbf{A} = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ a_1 & a_2 & \cdots & a_m \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} = \mathbf{U\Sigma V^T} = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ u_1 & u_2 & \cdots & u_n \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_m \\ \hline & & 0 & \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_m \end{bmatrix}$$

eigen faces

7. Singular Value Decomposition

Example



SVD: The fundamental theorem of multivariate data analysis. Because of its usefulness, the singular value decomposition is a fundamental technique for multivariate data analysis.

** Low-rank approximations*

** Principal components analysis*

